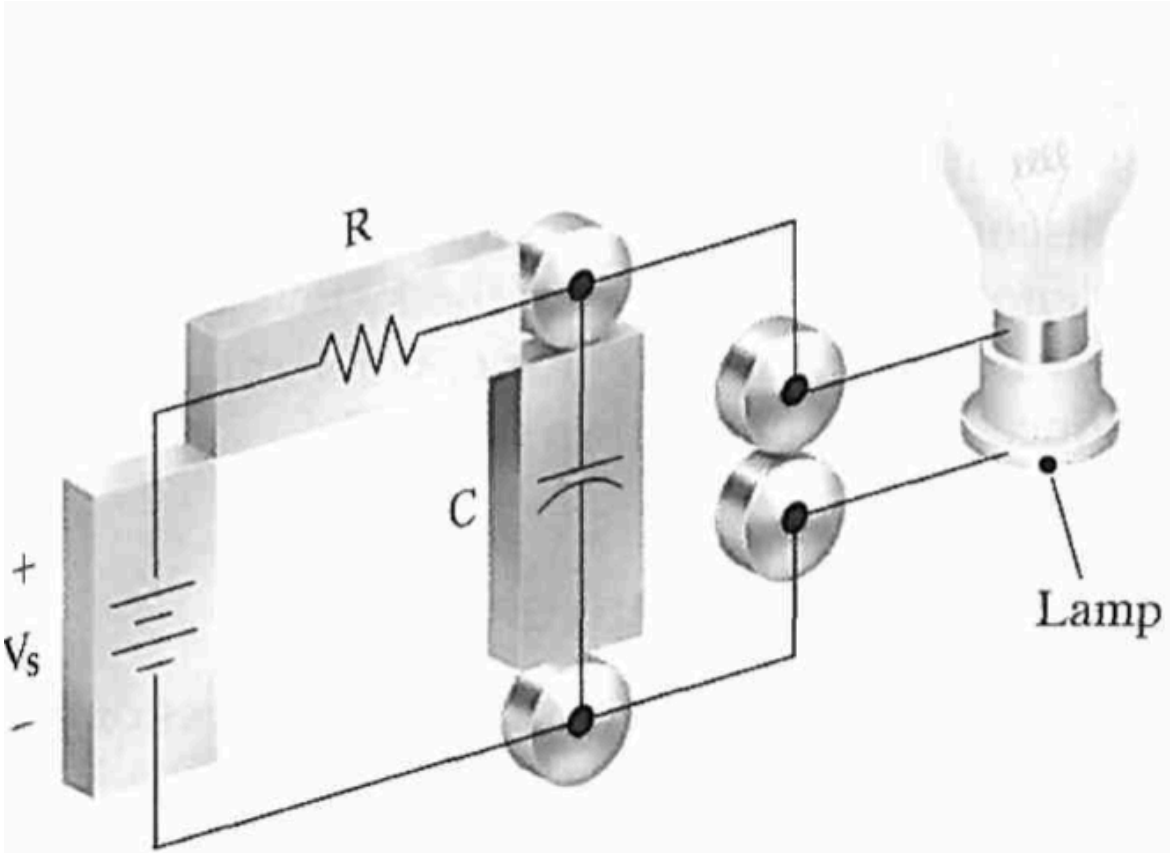
Response of First-Order *RL* and *RC* Circuits

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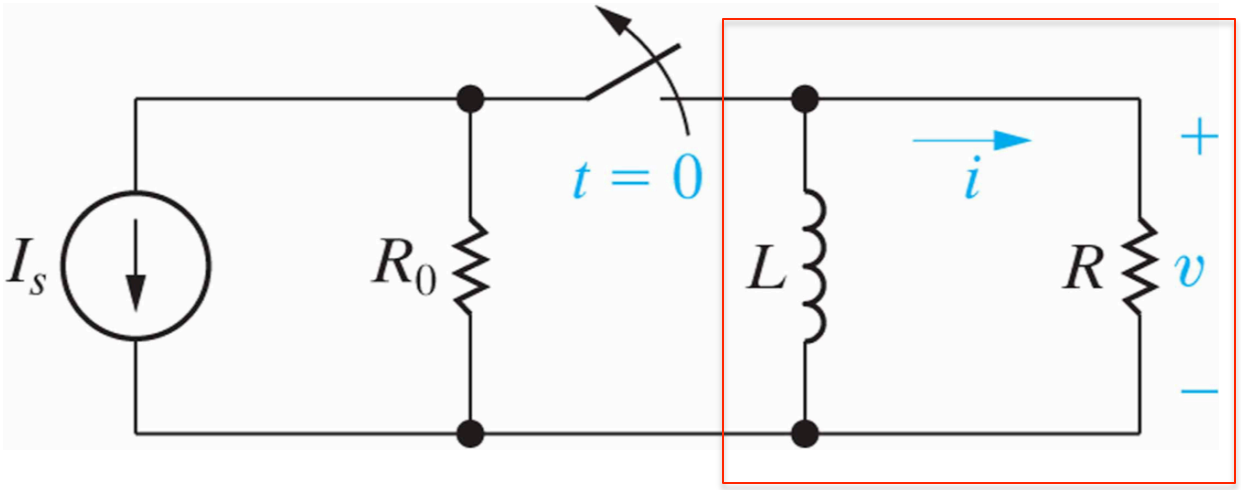
Structure

- The Natural Response of an *RL* Circuit
- The Natural Response of an *RC* Circuit
- The Step Response of *RL* and *RC* Circuits
- A General Solution for Step and Natural Responses
- Sequential Switching
- Unbounded Response
- The Integrating Amplifier

A Flashing Light Circuit



The Natural Response of an *RL* Circuit



a) The switch has been in a closed position for a long time: *all currents and voltages have reached a constant value*.

b) The inductor appears as a short circuit ($Ldi/dt = 0$) prior to the release of the stored energy.

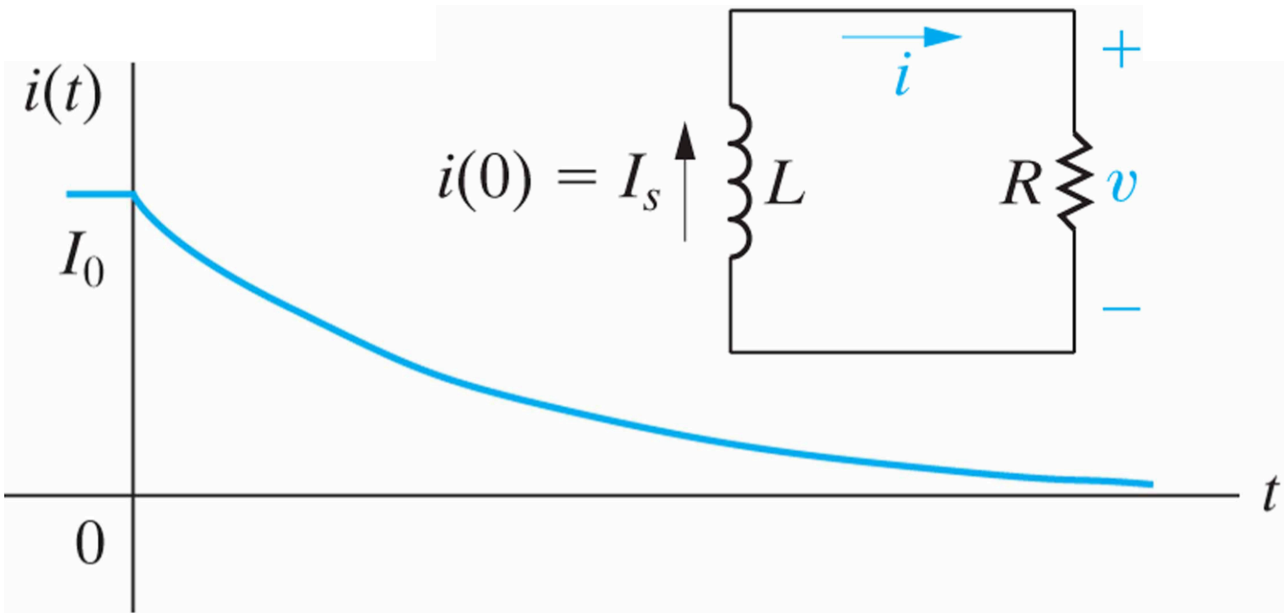
Expression for the Current

$$L\frac{di}{dt} + Ri = 0$$

$$\frac{di}{i} = -\frac{R}{L}dt$$

$$\int_{i(t_0)}^{i(t)} \frac{dx}{x} = -\frac{R}{L}\int_{t_0}^{t} dy \quad \Rightarrow \quad \ln\frac{i(t)}{i(0)} = -\frac{R}{L}t$$

$$\Rightarrow \quad i(t) = i(0)e^{-(R/L)t}$$



Expression for the Power and Energy

$$v = iR = I_0 R e^{-(R/L)t}, \qquad t \geq 0^+$$

$$p = vi$$

$$p = I_0^2 R e^{-2(R/L)t}$$

$$w = \int_0^t p\,dx = \int_0^t I_0^2 R e^{-2(R/L)x}\,dx$$

$$= \frac{1}{2(R/L)} I_0^2 R(1 - e^{-2(R/L)t})$$

$$= \frac{1}{2} L I_0^2 (1 - e^{-2(R/L)t}), \qquad t \geq 0.$$

The Significance of the Time Constant

$$i(t) = i(0)e^{-(R/L)t} \quad \Rightarrow \quad \tau = \text{time constant} = \frac{L}{R}$$

$$i(t) = I_0 e^{-t/\tau}, \qquad t \geq 0,$$

$$v(t) = I_0 R e^{-t/\tau}, \qquad t \geq 0^+,$$

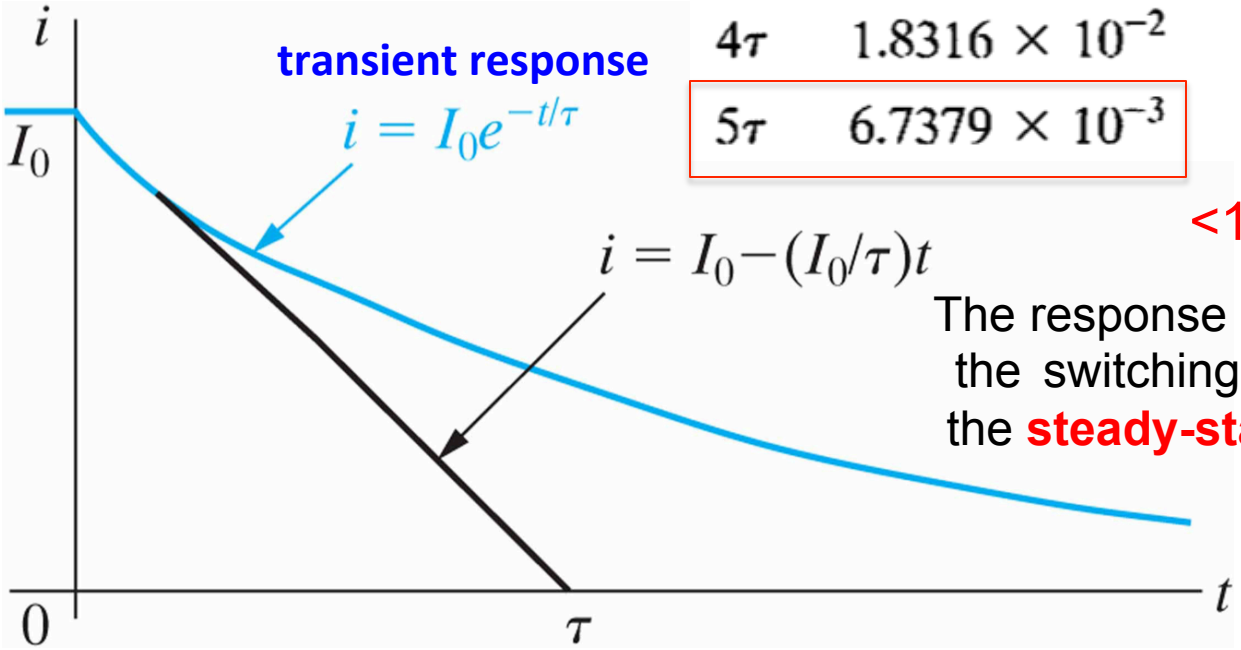
$$p = I_0^2 R e^{-2t/\tau}, \qquad t \geq 0^+,$$

$$w = \frac{1}{2} L I_0^2 (1 - e^{-2t/\tau}), \qquad t \geq 0.$$

The time constant is an important parameter for first-order circuits, and *it is convenient to think of the time elapsed after switching in terms of integral multiples of* τ: one time constant after the inductor has begun to release its stored energy to the resistor, the current has been reduced to e^{-1} , or approximately 0.37 of its initial value.

Steady-State Response

t	$e^{-t/\tau}$	t	$e^{-t/\tau}$
τ	3.6788×10^{-1}	6τ	2.4788×10^{-3}
2τ	1.3534×10^{-1}	7τ	9.1188×10^{-4}
3τ	4.9787×10^{-2}	8τ	3.3546×10^{-4}
4τ	1.8316×10^{-2}	9τ	1.2341×10^{-4}
5τ	6.7379×10^{-3}	10τ	4.5400×10^{-5}



<1%

The response that exists *a long time* after the switching has taken place is called the **steady-state response**

Calculating the natural response of an *RL* circuit can be summarized as follows:

1. Find the initial current, I_0, through the inductor.
2. Find the time constant of the circuit, $\tau = L/R$.
3. Use $I_0 e^{-t/\tau}$ to generate $i(t)$ from I_0 and τ.

Example #1

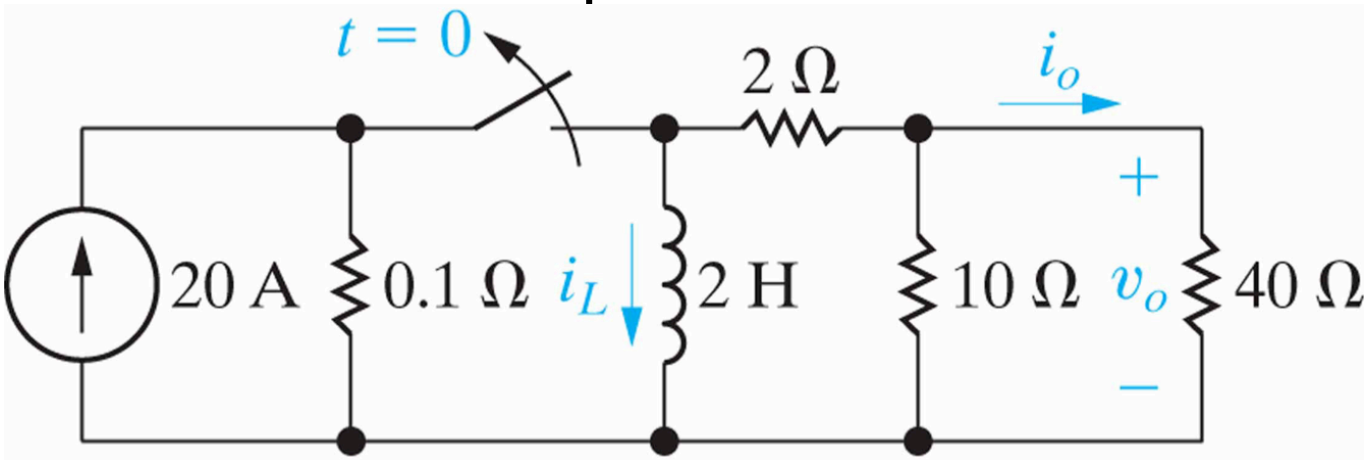
The switch in the circuit shown in Figure has been closed for a long time before it is opened at $t = 0$. Find

a) $i_L(t)$ for $t \geq 0$,

b) $i_o(t)$ for $t \geq 0^+$,

c) $v_o(t)$ for $t \geq 0^+$,

d) the percentage of the total energy stored in the 2 H inductor that is dissipated in the 10 Ω resistor.



Solution for Example #1

$$R_{eq} = 2 + (40 \parallel 10) = 10\,\Omega$$

$i_L(0^+)=20\text{A}$, $\tau = L/R_{eq} = 0.2$ s

a)

$$i_L(t) = 20e^{-5t}\,\text{A}, \quad t \geq 0$$

$$i_o = -i_L \frac{10}{10 + 40}$$

b)

$$v_o(t) = 40i_o = -160e^{-5t}\,\text{V}, \quad t \geq 0^+$$

c)

$$i_o(t) = -4e^{-5t}\,\text{A}, \quad t \geq 0^+$$

d)

$$p_{10\Omega}(t) = \frac{v_o^2}{10} = 2560e^{-10t}\,\text{W}, \qquad t \geq 0^+$$

$$w_{10\Omega}(t) = \int_0^\infty 2560e^{-10t}\,dt = 256\,\text{J}$$

$$w(0) = \frac{1}{2}Li^2(0) = \frac{1}{2}(2)(400) = 400\,\text{J}$$

$$\frac{256}{400}(100) = 64\%$$

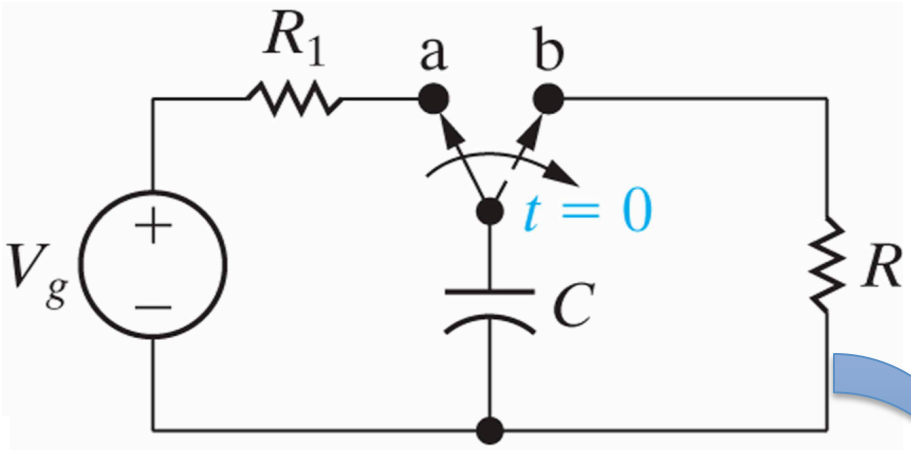
The Natural Response of an *RC* Circuit

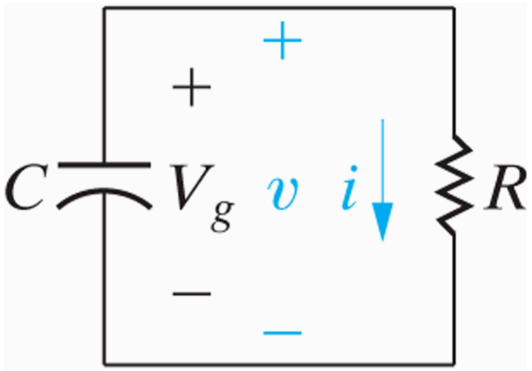
$$C\frac{dv}{dt} + \frac{v}{R} = 0$$

$$v(t) = v(0)e^{-t/RC}, \quad t \geq 0$$

$$\tau = RC$$

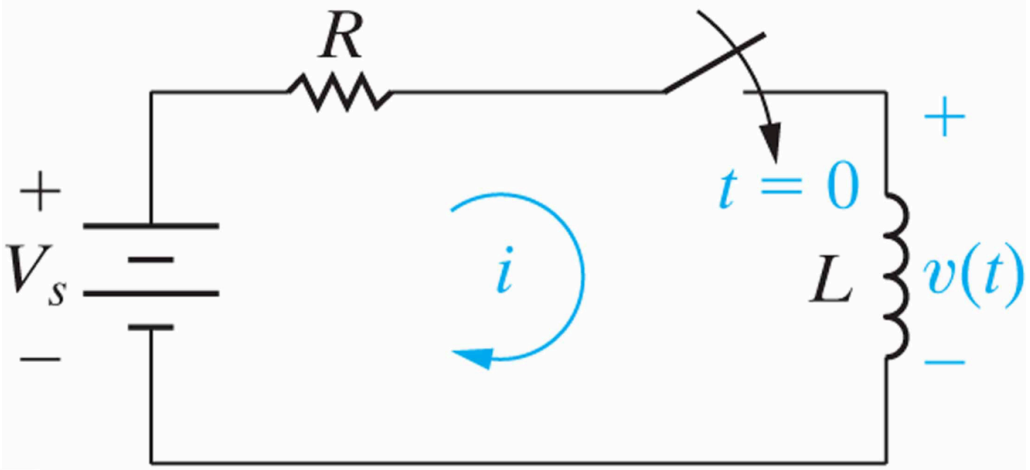
$$v(t) = V_0 e^{-t/\tau}, \quad t \geq 0$$





The Step Response of an *RL* Circuit

Kirchhoff's voltage law



$$V_s = Ri + L\frac{di}{dt}$$

$$\frac{di}{dt} = \frac{-Ri + V_s}{L} = \frac{-R}{L}\left(i - \frac{V_s}{R}\right)$$

$$\frac{di}{i - (V_s/R)} = \frac{-R}{L}dt \quad \Rightarrow \quad \int_{I_0}^{i(t)} \frac{dx}{x - (V_s/R)} = \frac{-R}{L}\int_0^t dy$$

$$\ln \frac{i(t) - (V_s/R)}{I_0 - (V_s/R)} = \frac{-R}{L}t$$

$$\frac{i(t) - (V_s/R)}{I_0 - (V_s/R)} = e^{-(R/L)t}$$

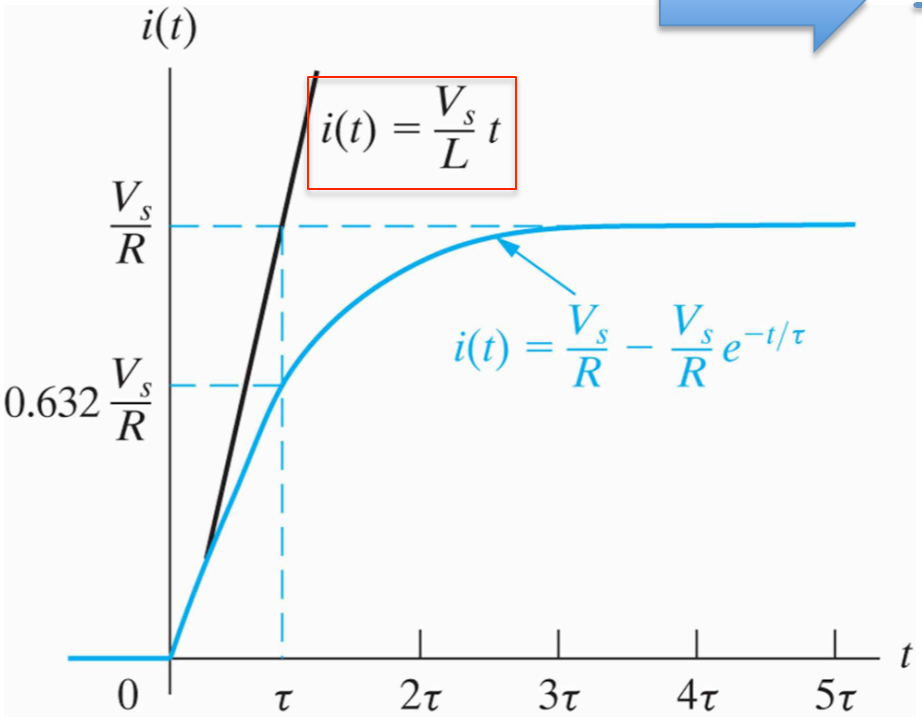
$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-(R/L)t}$$

$I_0 = 0$

$$i(t) = \frac{V_s}{R} - \frac{V_s}{R}e^{-(R/L)t}$$

Step Response Curve

$$i(t) = \frac{V_s}{R} - \frac{V_s}{R}e^{-(R/L)t}$$



$$i(\tau) = \frac{V_s}{R} - \frac{V_s}{R}e^{-1} \approx 0.6321\frac{V_s}{R}$$

$$\frac{di}{dt} = \frac{-V_s}{R}\left(\frac{-1}{\tau}\right)e^{-t/\tau} = \frac{V_s}{L}e^{-t/\tau}$$

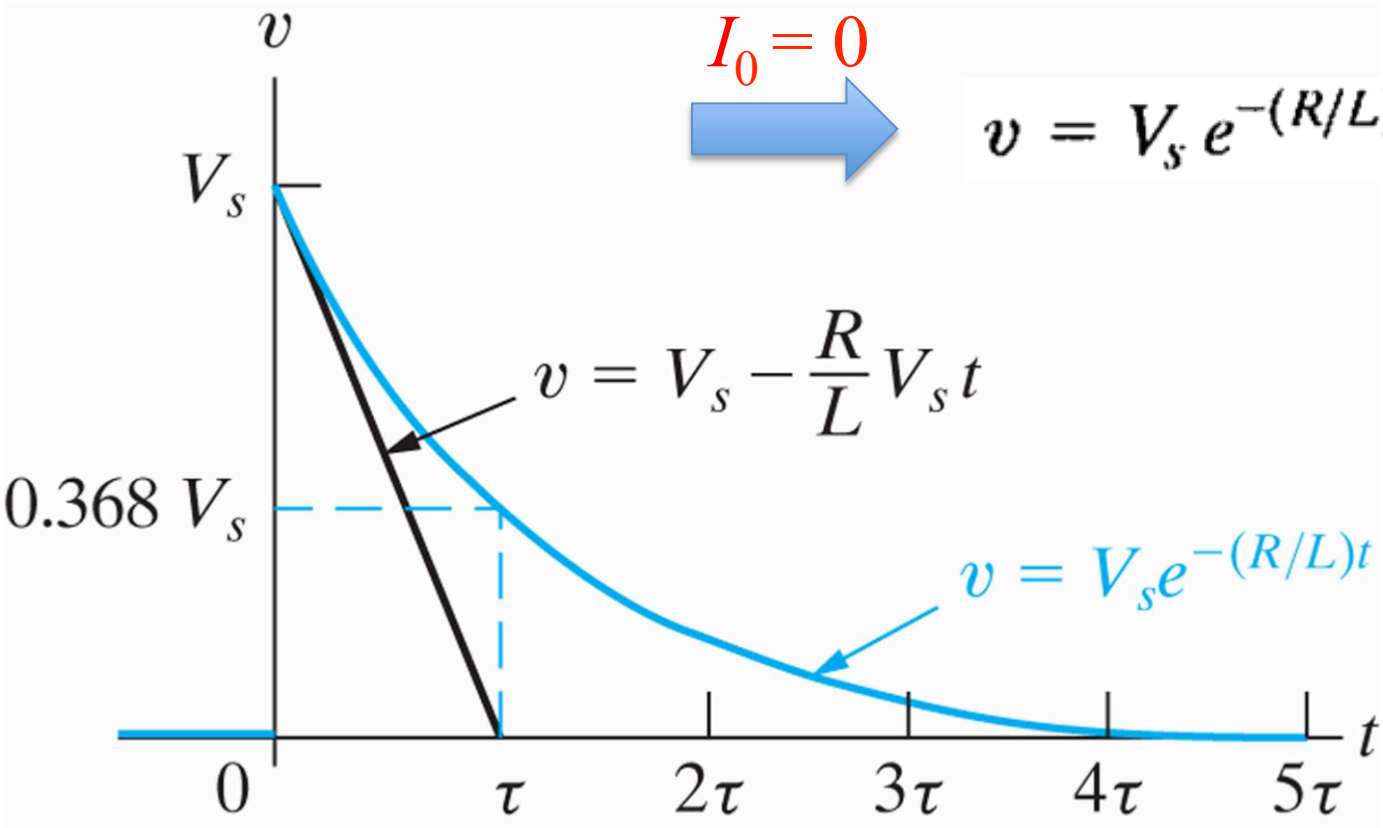
$$\frac{di}{dt}(0) = \frac{V_s}{L}$$

$$i = \frac{V_s}{L}\frac{L}{R} = \frac{V_s}{R}$$

τ

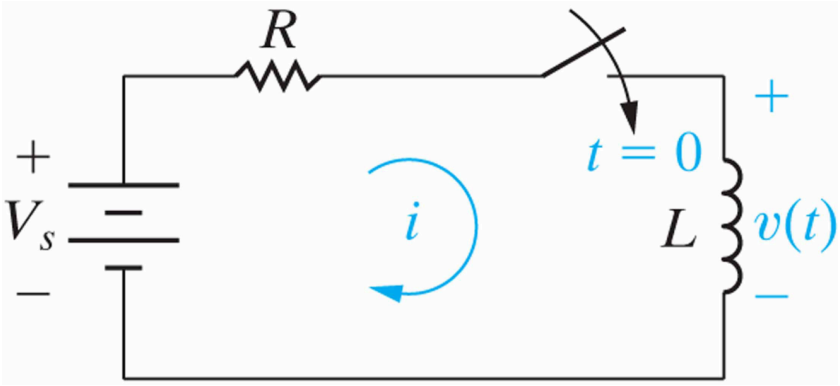
The voltage across an inductor is $L di/dt$, so for $t > 0^+$, we have:

$$v = L\left(\frac{-R}{L}\right)\left(I_0 - \frac{V_s}{R}\right)e^{-(R/L)t} = (V_s - I_0 R)e^{-(R/L)t}$$



Another Way to Get the Voltage Expression

$$i(t) = \frac{V_s}{R} - \frac{v(t)}{R}$$



$$\frac{di}{dt} = -\frac{1}{R}\frac{dv}{dt}$$

$$v = -\frac{L}{R}\frac{dv}{dt}$$

$$\frac{di}{dt} + \frac{R}{L}i = \frac{V_s}{L}$$

(From Eq. 7.29)

$$\frac{dv}{dt} + \frac{R}{L}v = 0$$

(Compare with Eq. 7.1)

Example #2

The switch in the circuit has been in position a for a long time. At $t = 0$, the switch moves from position a to position b. The switch is a make-before-break type; that is, *the connection at position b is established before the connection at position a is broken*, so there is no interruption of current through the inductor.

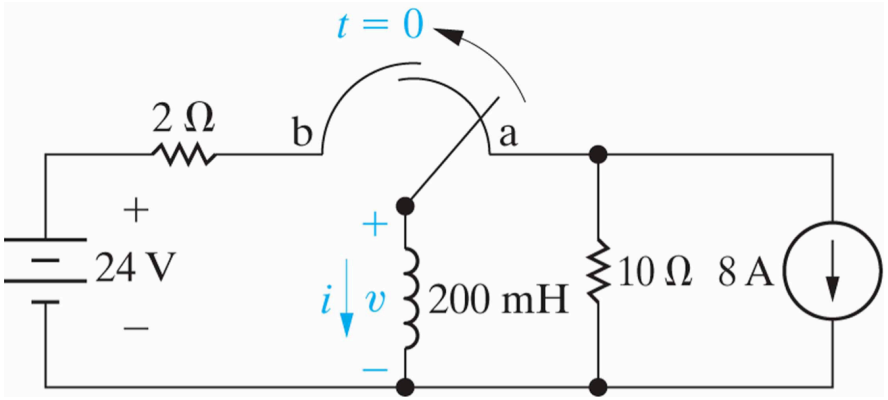
a) Find the expression for $i(t)$ for $t \geq 0$.

b) What is the initial voltage across the inductor just after the switch has been moved to position b?

c) Does this initial voltage make sense in terms of circuit behavior?

d) How many milliseconds after the switch has been moved does the inductor voltage equal 24 V?

e) Plot both $i(t)$ and $v(t)$ versus t.



Solution for Example #2

$I_0 = -8$ A $\tau = L/R = 0.2$ H / 2 Ω = 0.1 s

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-(R/L)t}$$

a) $$i = 12 + (-8 - 12)e^{-t/0.1} = 12 - 20e^{-10t}\ \text{A}, \quad t \geq 0$$

b) $$v = L\frac{di}{dt} = 0.2(200e^{-10t}) = 40e^{-10t}\ \text{V}, \quad t \geq 0^+$$

c) $$v(0^+) = 40\ \text{V}$$

Yes; in the instant after the switch has been moved to position b, the inductor sustains a current of 8 A counterclockwise around the newly formed closed path. *This current causes a 16 V drop across the 2 Ω resistor. This voltage drop adds to the drop across the source, producing a 40 V drop across the inductor.*

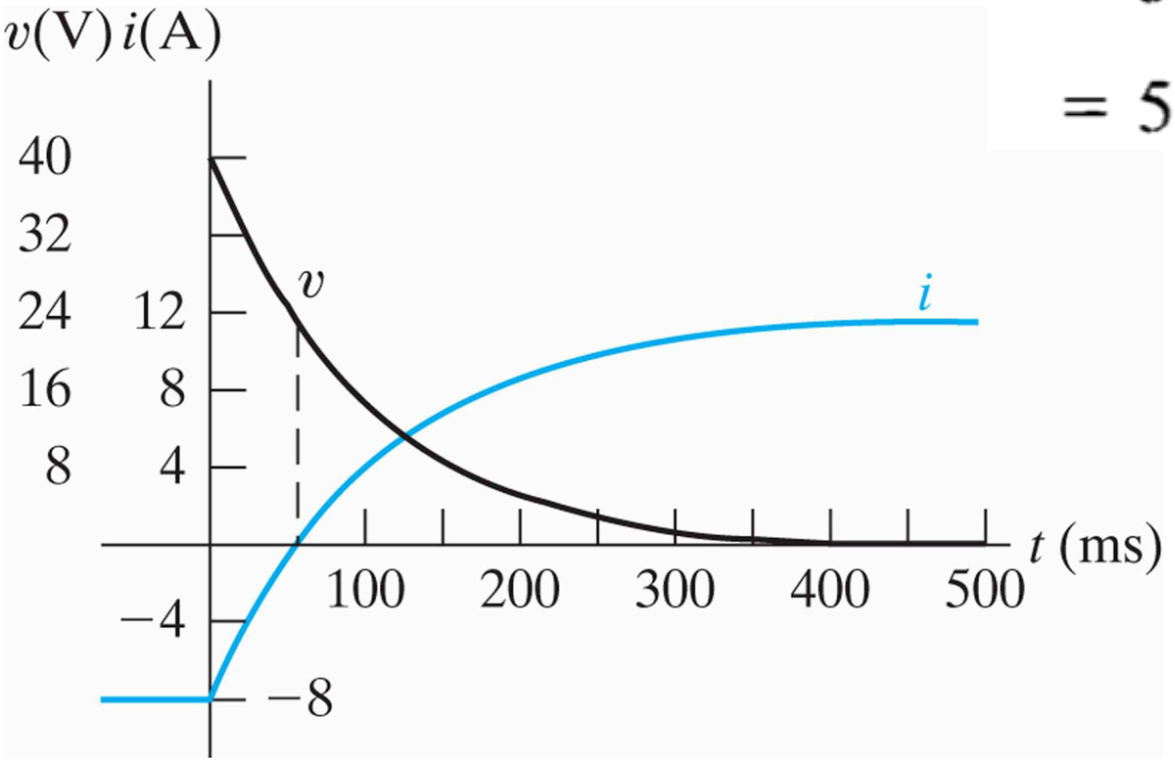
d) We find the time at which the inductor voltage equals 24 V by solving the expression

$$24 = 40e^{-10t} \quad \Rightarrow \quad t = \frac{1}{10}\ln\frac{40}{24}$$

$$= 51.08 \times 10^{-3}$$

$$= 51.08 \text{ ms.}$$

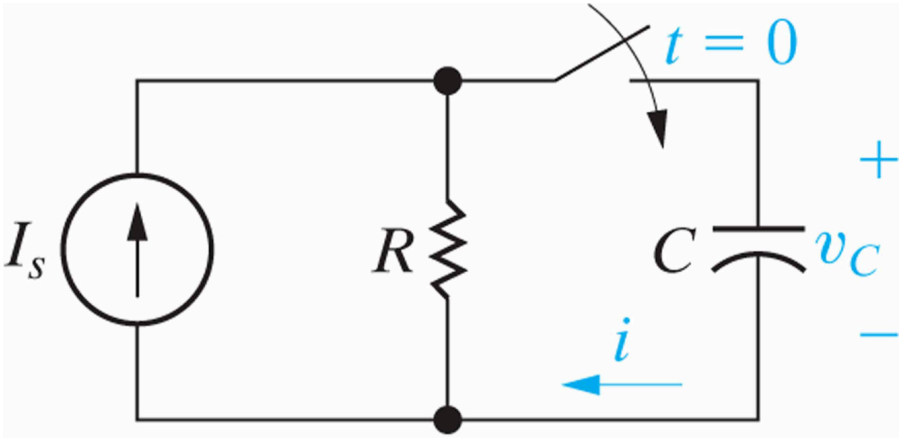
e)



The Step of an RC Circuit

$$C\frac{dv_C}{dt} + \frac{v_C}{R} = I_s$$

$$\frac{dv_C}{dt} + \frac{v_C}{RC} = \frac{I_s}{C}$$



By comparing with Eq. 7.48

$$v_C = I_sR + (V_0 - I_sR)e^{-t/RC}, \quad t \geq 0$$

$$\frac{di}{dt} + \frac{1}{RC}i = 0 \quad \Rightarrow \quad i = \left(I_s - \frac{V_0}{R}\right)e^{-t/RC}, \quad t \geq 0^+$$

A General Solution for Step and Natural Response

$$\frac{dx}{dt} + \frac{x}{\tau} = K \quad \Rightarrow \quad x_f = K\tau$$

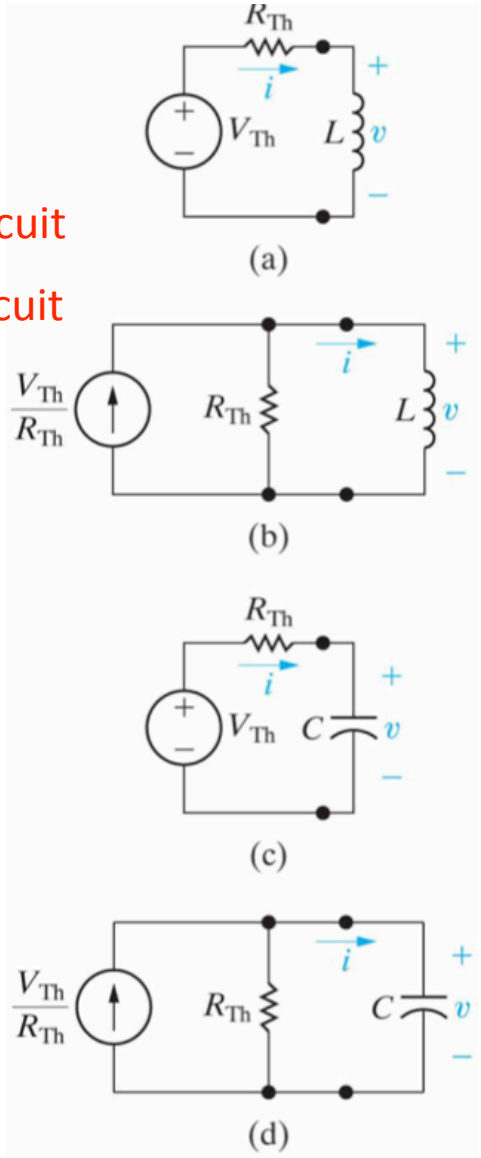
$\tau = L/R$ for RL circuit

$\tau = RC$ for RC circuit

$$\frac{dx}{dt} = \frac{-x}{\tau} + K = \frac{-(x - K\tau)}{\tau} = \frac{-(x - x_f)}{\tau}$$

$$\Rightarrow \quad \frac{dx}{x - x_f} = \frac{-1}{\tau} dt$$

$$\int_{x(t_0)}^{x(t)} \frac{du}{u - x_f} = -\frac{1}{\tau} \int_{t_0}^{t} dv \quad \Leftarrow$$



$$x(t) = x_f + [x(t_0) - x_f]e^{-(t-t_0)/\tau}$$

$$\begin{pmatrix}\text{the unknown}\\ \text{variable as a}\\ \text{function of time}\end{pmatrix} = \begin{pmatrix}\text{the final}\\ \text{value of the}\\ \text{variable}\end{pmatrix} + \left[\begin{pmatrix}\text{the initial}\\ \text{value of the}\\ \text{variable}\end{pmatrix} - \begin{pmatrix}\text{the final}\\ \text{value of the}\\ \text{variable}\end{pmatrix}\right] \times e^{\frac{-[t-(\text{time of switching})]}{(\text{time constant})}}$$

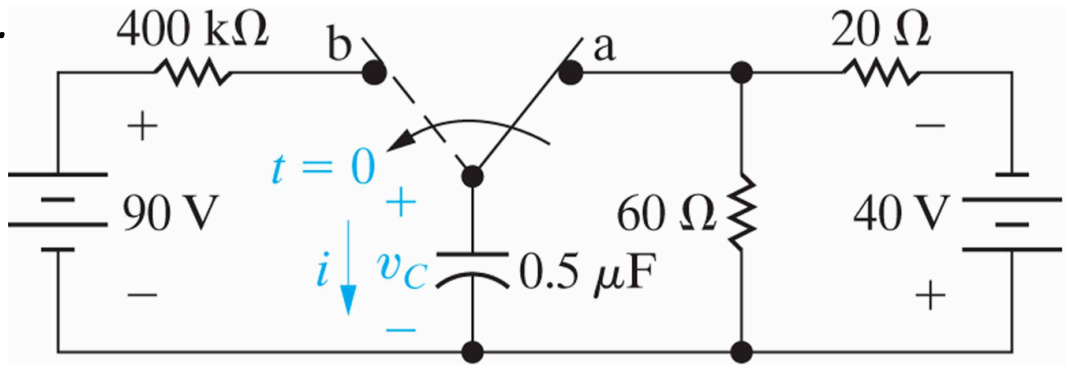
Helpful Notes

1. Identify the variable of interest for the circuit. For *RC* circuits, it is most convenient to choose the *capacitive voltage*; for *RL* circuits, it is best to choose the *inductive current*.
2. Determine the initial value of the variable, which is its value at t_0. Note that if you choose capacitive voltage or inductive current as your variable of interest, it is not necessary to distinguish between $t = t_0^-$ and $t = t_0^+$. This is because *they both are continuous variables*. If you choose another variable, you need to remember that its initial value is defined at $t = t_0^+$.
3. Calculate the *final value* of the variable, which is its value as $t \rightarrow \infty$.
4. Calculate the *time constant* for the circuit.

Example #3

The switch in the circuit has been in position a for a long time. At t = 0 the switch is moved to position b.

a) What is the initial value of v_C?
b) What is the final value of v_C?
c) What is the time constant of the circuit when the switch is in position b?
d) What is the expression for $v_C(t)$ when $t \geq 0$?
e) What is the expression for $i(t)$ when $t \geq 0^+$?
f) How long after the switch is in position b does the capacitor voltage equal zero?
g) Plot $v_C(t)$ and $i(t)$ versus t.



Solution for Example #3

a) $v_C(0) = -40\ \text{V} \times 60 / (20 + 60) = -30\ \text{V}$

b) $v_C(\infty) = 90\ \text{V}$

c) $\tau = \text{RC} = (400 \times 10^3)(0.5 \times 10^{-6}) = 0.2\ \text{s}$

d) $v_C(t) = v_C(\infty) + [v_C(0) - v_C(\infty)]\, e^{-t/\tau} = 90 - 120e^{-5t}\ \text{V},\ t \geq 0$

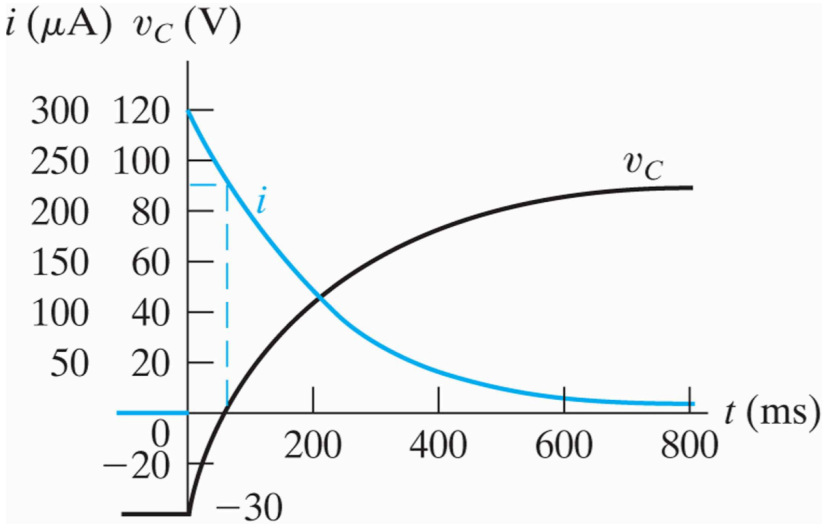
e) $i(t) = i(\infty) + [i(0) - i(\infty)]\, e^{-t/\tau} = 300e^{-5t}\ \mu\text{A},\ t \geq 0$

$v_C(t) = 0$

$i(\infty) = [90 - v_C(0)]\ \text{V} / 400\ \text{k}\Omega = 300\ \mu\text{A}$

f) $t = 0.2 \ln(120/90) = 57.54\ \text{ms}$

g)



Sequential Switching

- Definition: switching occurs more than once in a circuit.
- Method: derive the expressions for $v(t)$ and $i(t)$ for a given position of the switch or switches and then use these solutions to determine the initial conditions for the next position of the switch or switches.
- Note: anything but inductive currents and capacitive voltages can change instantaneously at the time of switching.

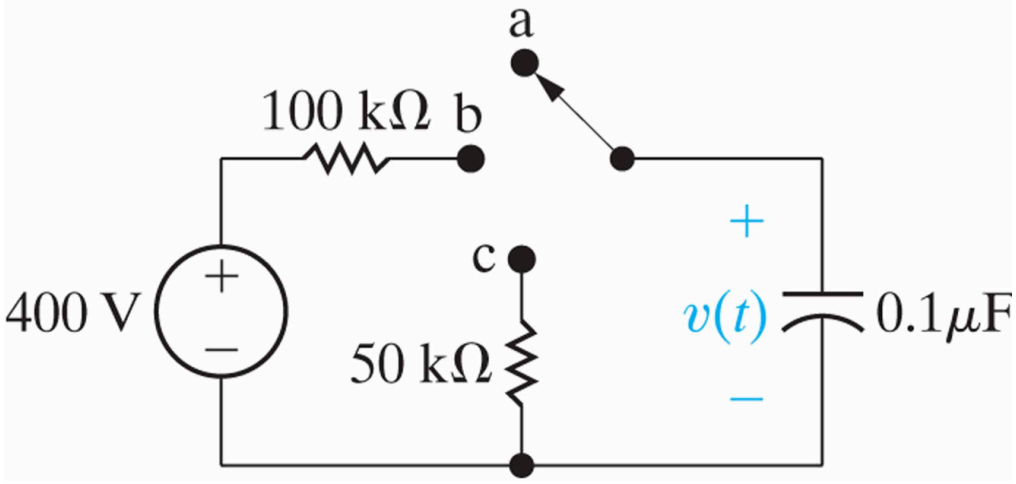
Example #4

The uncharged capacitor in the circuit is initially switched to terminal a of the three-position switch. At $t = 0$, the switch is moved to position b, where it remains for 15 ms. After the 15 ms delay, the switch is moved to position c, where it remains indefinitely.

a) Derive the numerical expression for the voltage across the capacitor.

b) Plot the capacitor voltage versus time.

c) When will the voltage on the capacitor equal 200 V?



Solution for Example #4

a)

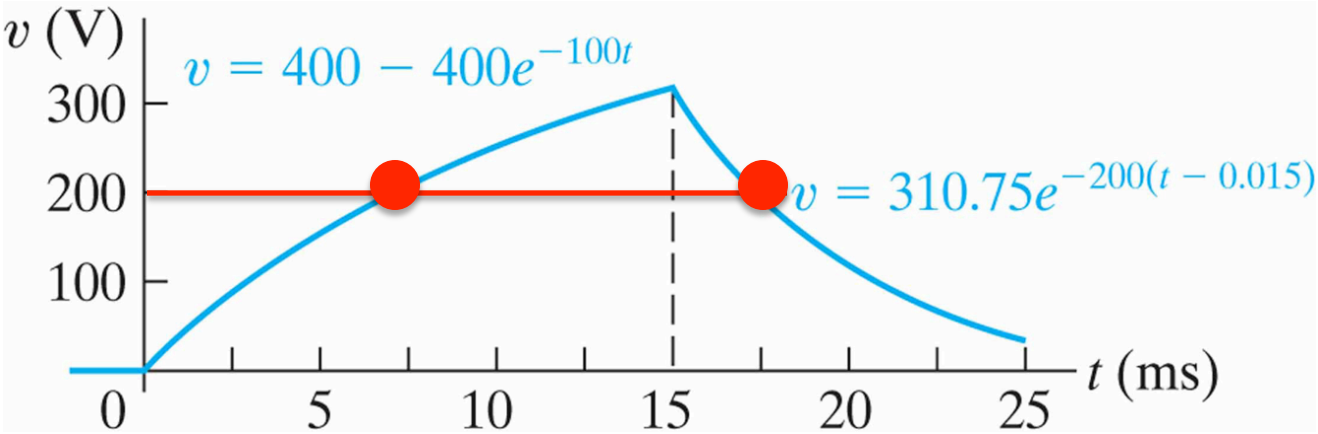
Touch b: $v(0) = 0$ V; $v(\infty) = 400$ V; $\tau = RC = 10$ ms

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau} = 400 - 400e^{-100t}\ \text{V},\ 0 \le t \le 15\ \text{ms}$$

Touch c: $v(0.015) = 400(1-e^{-1.5}) = 310.75$ V; $v(\infty) = 0$ V; $\tau = RC = 5$ ms

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-(t-0.015)/\tau} = 310.75e^{-200(t-0.015)}\ \text{V},\ t \ge 15\ \text{ms}$$

b)



c) $t_1 = 6.93$ ms; $t_2 = 17.20$ ms

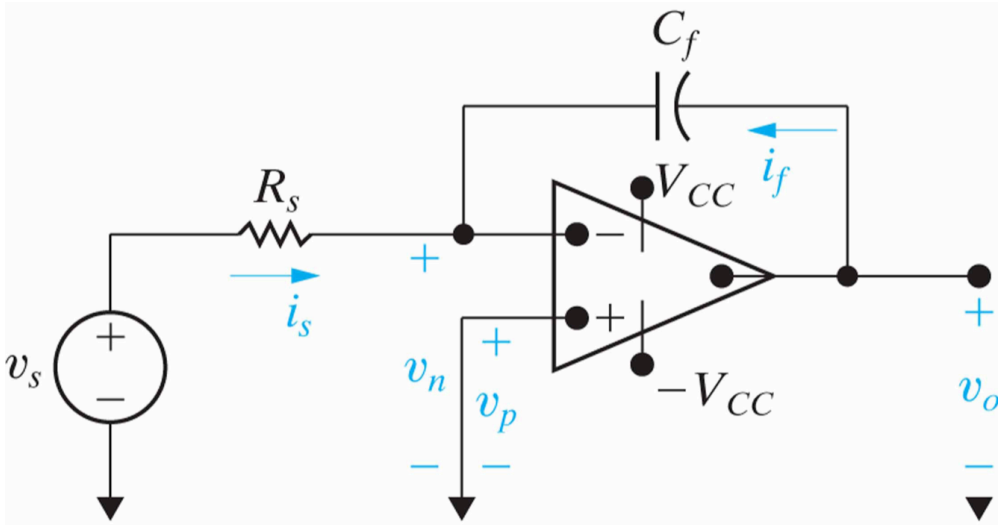
The Integrating Amplifier

$v_n = v_p = 0$

$$i_s = \frac{v_s}{R_s}$$

$$i_f = C_f \frac{dv_o}{dt}$$

$$i_f + i_s = 0$$



$$\frac{dv_o}{dt} = -\frac{1}{R_s C_f} v_s$$

$$v_o(t) = -\frac{1}{R_s C_f} \int_{t_0}^{t} v_s \, dy + v_o(t_0)$$

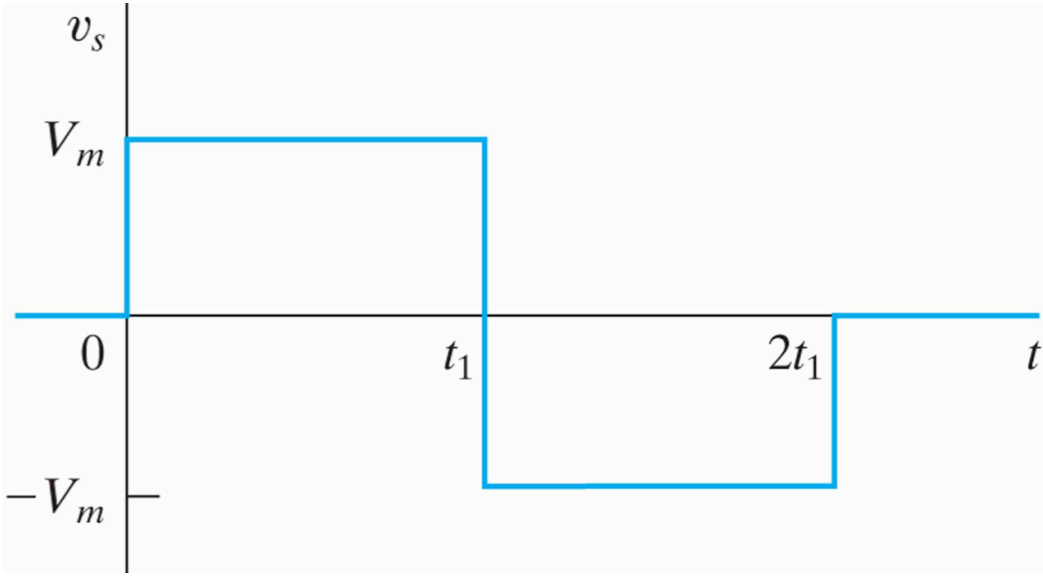
$$= 0$$

$$v_o(t) = -\frac{1}{R_s C_f} \int_{t_0}^{t} v_s \, dy$$

Example #5

Assume that the numerical values for the signal voltage are $V_m = 50$ mV and $t_1 = 1$ s. This signal voltage is applied to the integrating-amplifier circuit shown in Fig. 7.40. The circuit parameters of the amplifier are $R_s = 100$ kΩ, $C_f = 0.1$ μF, and $V_{CC} = 6$ V. The initial voltage on the capacitor is zero.

a) Calculate $v_o(t)$.
b) Plot $v_o(t)$ versus t.



Solution for Example #5

a)

$$v_o(t) = -\frac{1}{R_s C_f}\int_{t_0}^{t} v_s\, dy$$

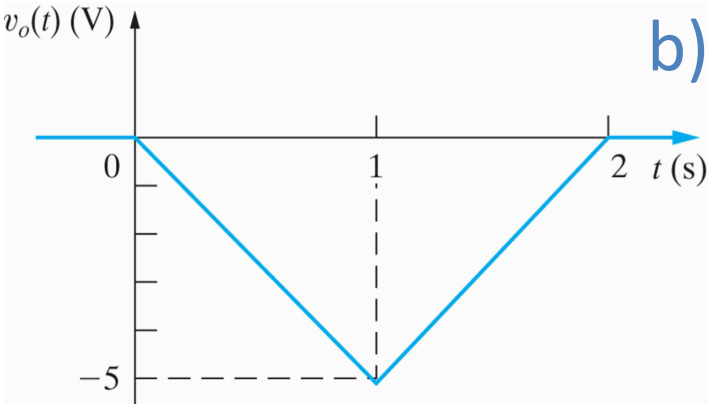
$$v_o = -\frac{1}{R_s C_f} V_m t + 0, \qquad 0 \le t \le t_1$$

$$v_o = -\frac{1}{R_s C_f}\int_{t_1}^{t} (-V_m)\, dy - \frac{1}{R_s C_f} V_m t_1 = \frac{V_m}{R_s C_f} t - \frac{2V_m}{R_s C_f} t_1, \qquad t_1 \le t \le 2t_1$$

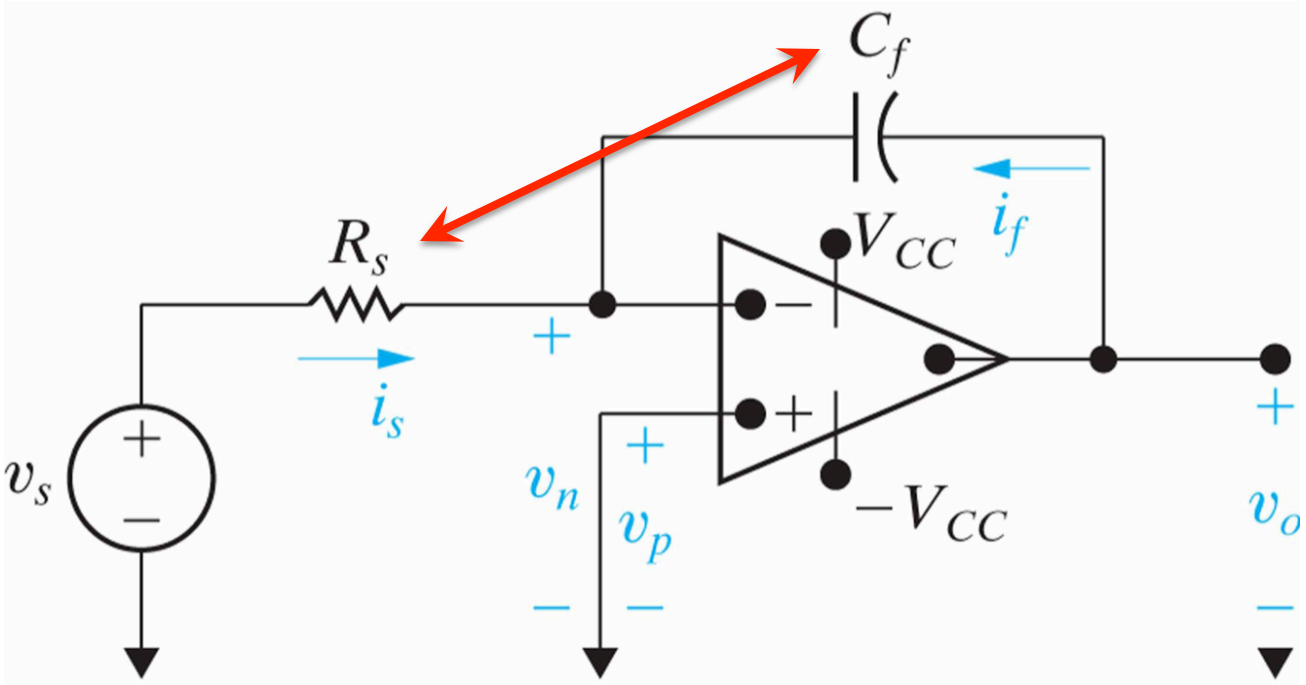
$$v_o = \frac{-1}{(100 \times 10^3)(0.1 \times 10^{-6})} 50 \times 10^{-3} t + 0 = -5t \text{ V}, \qquad 0 \le t \le 1 \text{ s}$$

$$v_o = (5t - 10) \text{ V} \qquad 1 \le t \le 2 \text{ s}$$

b)

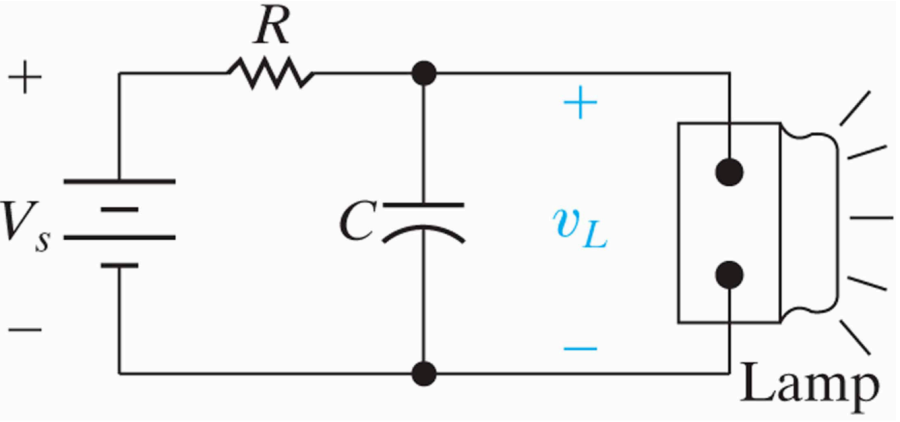


The Differentiating Amplifier

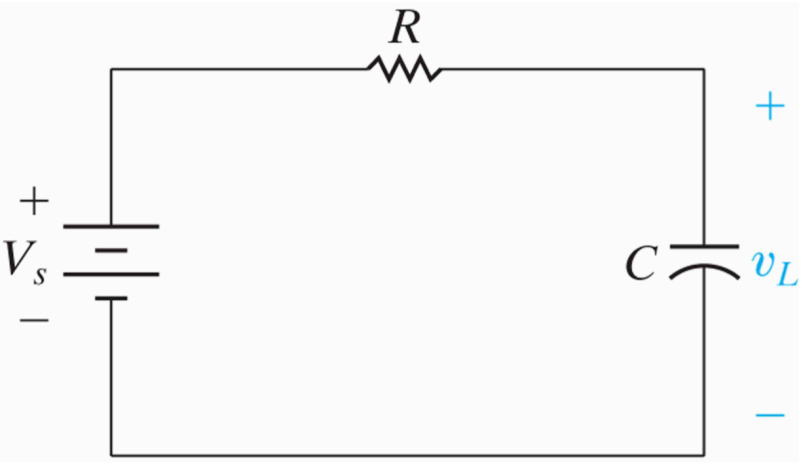


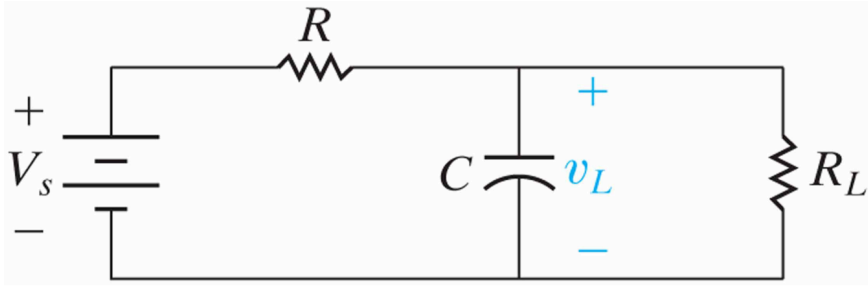
$$v_o = -R_s C_f \frac{dv_s}{dt}$$

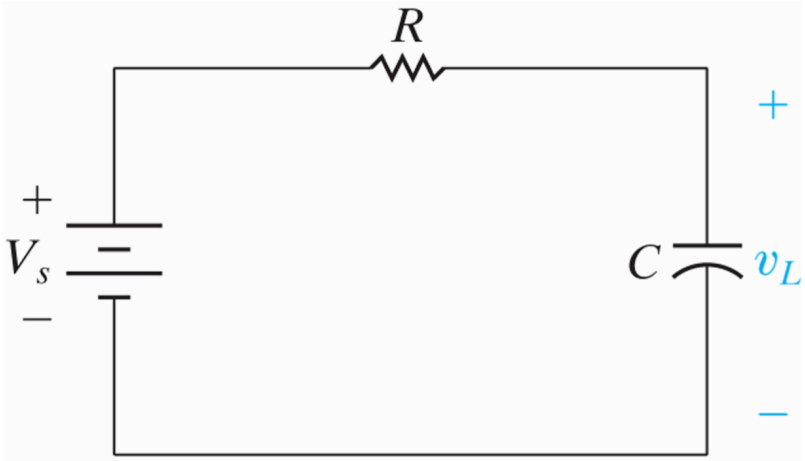
A Flashing Light Circuit



The lamp in this circuit starts to conduct whenever the lamp voltage reaches a value $V_{\max}$. During the time the lamp conducts, it can be modeled as a resistor whose resistance is R_L. The lamp will continue to conduct until the lamp voltage drops to the value $V_{\min}$. When the lamp is not conducting, it behaves as an open circuit.





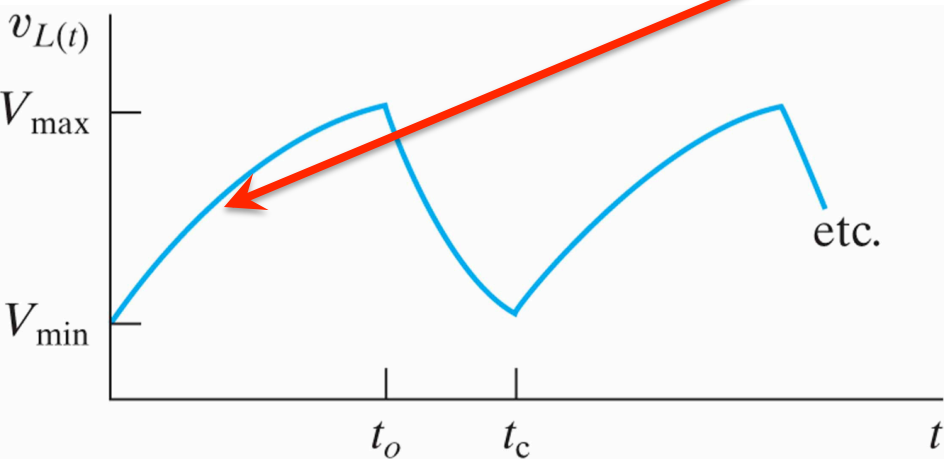


$$v_L(0) = V_{\min}$$

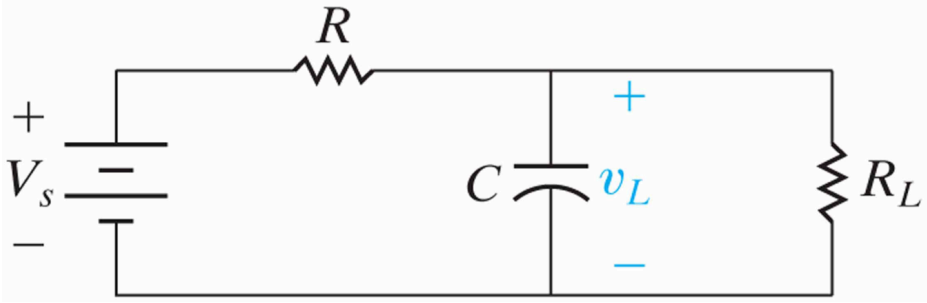
$$v_L(\infty) = V_s$$

$$\tau = RC$$

$$v_L(t) = V_s + (V_{\min} - V_s)e^{-t/RC}$$



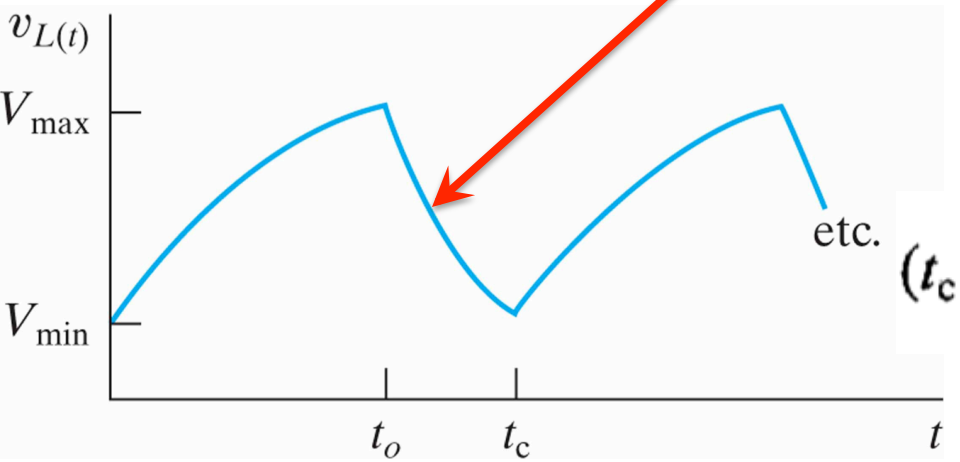
$$t_o = RC \ln \frac{V_{\min} - V_s}{V_{\max} - V_s}$$



$$V_{\text{Th}} = \frac{R_L}{R + R_L} V_s$$

$$\tau = \frac{RR_LC}{R + R_L}$$

$$v_L(t) = V_{\text{Th}} + (V_{\text{max}} - V_{\text{Th}})e^{-(t-t_o)/\tau}$$



$$(t_c - t_o) = \frac{RR_LC}{R + R_L} \ln \frac{V_{\text{max}} - V_{\text{Th}}}{V_{\text{min}} - V_{\text{Th}}}$$

Summary

- Natural response for RL and RC circuits
- Time constants RL and RC circuits (L/R, RC)
- Step response for RL and RC circuits
- Sequential switching
- Unbounded response
- Integrating and differentiating amplifier